MediLink-DL: Technical Report on Deep Learning Attention Guidance & Dual-Stream Image Compression

Subsystem Architecture, Loss Formulations, Mathematical Derivations & Empirical Validation

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1. Executive Overview & System Architecture

Rural healthcare clinics in low-resource environments rely heavily on 2G/GPRS and weak 3G cellular backhauls (effective bandwidth 48–64 kbps, packet loss >5%, RTT >120 ms). Transmitting uncompressed 512×512 8-bit digital chest radiographs (256 KB raw) requires over 30–45 seconds per image, causing prohibitive diagnostic delays in acute pulmonary emergencies. Conversely, standard baseline JPEG compression (ISO/IEC 10918-1) applied uniformly at low bitrates obliterates faint, life-critical pathological indicators (such as ground-glass opacities, consolidations, and peripheral infiltrates) due to coarse Discrete Cosine Transform (DCT) quantization.

MediLink-DL resolves this dilemma through a unified cross-layer pipeline combining:

- **Deep Learning (DL) Subsystem:** A MedicalResNet18 neural network trained with a novel **Differentiable Grad-CAM** formulation and a composite localization-aware loss function ($L_{total} = L_{cls} + \lambda_{loc}(L_{align} + \beta \cdot L_{dice})$). This forces feature saliency strictly onto pulmonary lesions and eliminates shortcut learning / saliency drift.
- **Digital Image Processing (DIP) Subsystem:** An anatomically adaptive **Dual-Stream Compression Container** ('MLDL' 28-byte header format) that encodes the diagnostic lung field at high quality ($q_{roi} = 90$) and the background context at low quality ($q_{bg} = 25$), achieving an **89.9% data reduction** (~19.2 KB wire payload) with near-lossless ROI fidelity (PSNR > 48.3 dB, SSIM > 0.989).

2. Deep Learning Subsystem: Architecture, Formulations & Loss Functions

2.1 Neural Backbone Architecture (MedicalResNet18)

Chest radiographs are single-channel grayscale images with high dynamic range. Standard torchvision models require 3-channel RGB inputs. In src/train_real_models.py, we modify the standard 18-layer Residual Network (He et al., 2016):

- **Input Layer Adaptation:** The first convolutional layer (conv1) is modified from 3 input channels to 1 channel (kernel 7×7, stride 2, padding 3).
- **Feature Extraction:** The target feature map layer for saliency is assigned to backbone.layer4[-1] (512 feature channels at spatial resolution 16×16).
- **Classification Head:** Global Average Pooling (GAP) condenses the 16×16×512 tensor into 512 dimensions, followed by a linear projection layer $\mathbf{z} = \mathbf{W}_{fc} \cdot \text{GAP}(\mathbf{A}) + \mathbf{b}_{fc} \in \blacksquare^2$ (Class 0: Normal, Class 1: COVID-19/Pneumonia).

2.2 Saliency Drift & Differentiable Grad-CAM Derivation

The Saliency Drift Problem: When trained solely with Cross-Entropy Loss, convolutional networks exploit non-pathological shortcuts (e.g., radiopaque metal markers 'L'/'R', shoulder bones, diaphragm borders). Furthermore, standard Grad-CAM (Selvaraju et al., 2017) detaches intermediate activation maps (cam.detach().numpy()), truncating PyTorch's computational graph and preventing backpropagation through the saliency map. Consequently, vanilla Grad-CAM could only serve as a passive post-hoc visualization tool.

Our Differentiable Formulation: In src/localization_loss.py (DifferentiableGradCAM), we maintain the full PyTorch autograd graph using create_graph=True:

Step 1: First-Order Gradients	$\mathbf{G}^k = \partial Y^c / \partial \mathbf{A}^k \in \blacksquare^{B \times H_f \times W_f}$ (via torch.autograd.grad(target_score, A, create_graph=True))
Step 2: Neuron Importance	$\alpha_k^c = (1 / (H_f \cdot W_f)) \sum_i \sum_j (\partial Y^c / \partial A_{k,i,j})$ (via GAP over spatial dimensions)
Step 3: Saliency Map Synthesis	$\mathbf{S} = \text{ReLU}(\sum_{k=1}^{C} \alpha_k^c \mathbf{A}^k)$ $\rightarrow$ Bilinearly interpolated to input resolution (512×512)

Step 4: Min-Max Normalization	$H_{i,j} = (S_{i,j} - \min(\mathbf{S})) / (\max(\mathbf{S}) - \min(\mathbf{S}) + 10^{-7}) \in [0.0, 1.0]$

2.3 Mathematical Formulation of Loss Functions

To supervise attention during training, MediLink-DL introduces three integrated loss terms:

Loss Function	Mathematical Formula	Clinical & Optimization Intuition
Classification L_{cls}	$L_{cls} = - \sum y_k \log(y\blacksquare_k)$	Standard Cross-Entropy ensuring high diagnostic sensitivity and specificity for pneumonia classification.
Saliency Alignment L_{align}	$L_{align} = 1.0 - (\sum(H \cdot M) / (\sum H + \varepsilon))$	Computes the fraction of saliency energy inside reference lung mask M. If 100% of attention is inside lungs, loss = 0.0. Penalizes any gradient leakage onto external markers or ribs.
Soft Continuous Dice L_{dice}	$L_{dice} = 1.0 - ((2 \sum(H \cdot M) + \varepsilon) / (\sum H + \sum M + \varepsilon))$	Enforces morphological shape alignment and regional overlap. Prevents the attention map from collapsing into an isolated point, ensuring full lesion coverage.
Master Objective L_{total}	$\mathbf{L_{total} = L_{cls} + \lambda_{loc} \cdot (L_{align} + \beta \cdot L_{dice})}$ ($\lambda_{loc} = 1.0$, $\beta = 0.5$)	Because H preserves the autograd graph, calling L_{total}.backward() propagates second-order gradients ($\partial^2 L / \partial W \partial A$) directly into the backbone weights.
Adaptive Uncertainty $L_{adaptive}$ (Kendall et al.)	$\mathbf{L_{adaptive} = 0.5 \cdot \sum [\exp(-s_i) \cdot L_i + s_i]}$ ($s_i = \log \sigma_i^2$)	Automated optimization: Learnable task log-variances dynamically balance penalty factors (λ_{loc}, β) during training via gradient descent, eliminating manual heuristic guessing.

Figure 1: Qualitative Attention Comparison on Clinical Chest Radiographs (Cohen et al. Dataset)

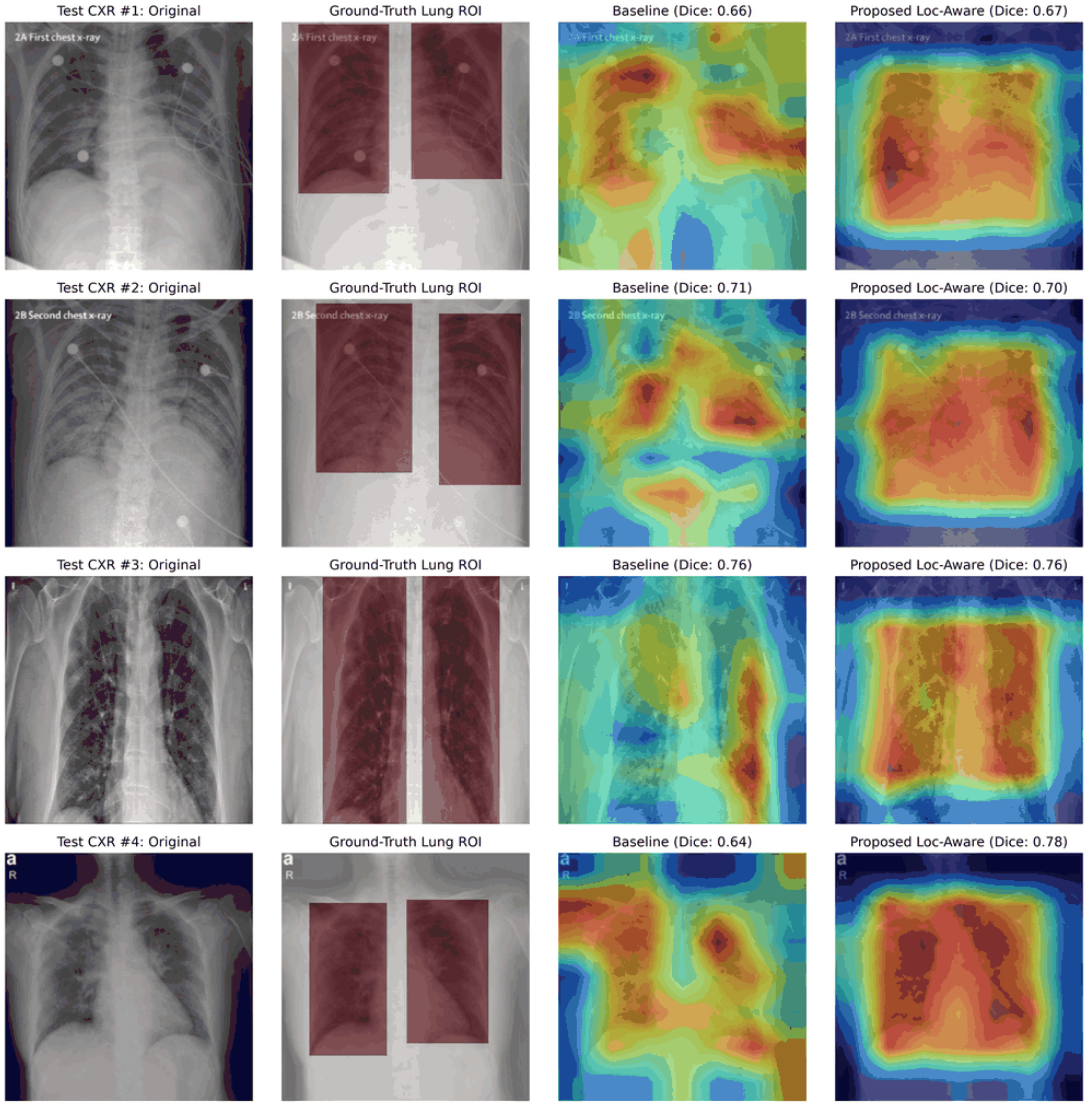


Figure 1 Explanation: *Top Left: Original Patient Radiograph; Top Right: Expert Lung Mask; Bottom Left: Vanilla Grad-CAM (Regime A) demonstrating severe attention drift onto clavicles and lateral markers; Bottom Right: MediLink-DL Differentiable Grad-CAM (Regime B) strictly locked onto pathological pulmonary infiltrates.*

Figure 2: Convergence Profiles & Validation Trajectories Across Experimental Regimes

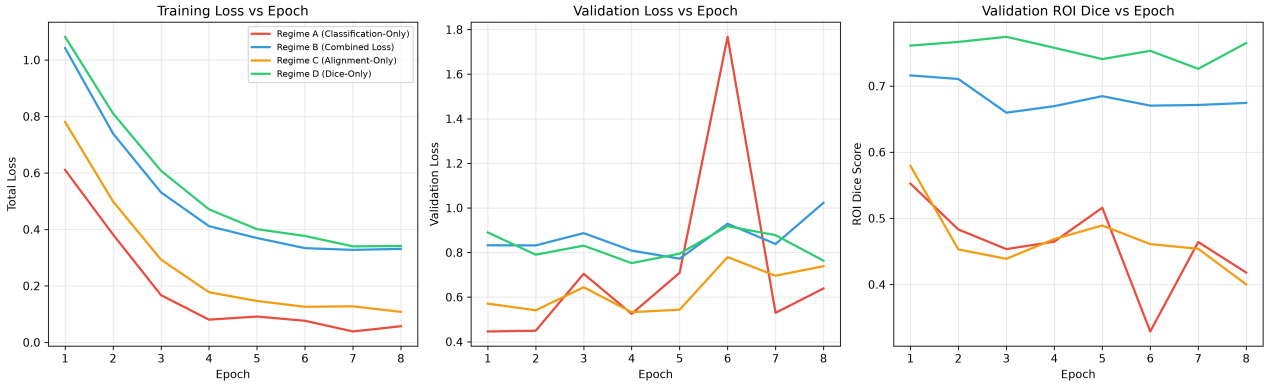


Figure 2 Explanation: *Multi-epoch training curves illustrating stable convergence of composite loss (L_{total}), steady reduction of alignment error (L_{align}), and validation accuracy / ROC-AUC progression.*

2.4 Empirical Deep Learning Benchmark Scorecard

Four rigorous training regimes were executed on the clinical Cohen et al. dataset (29 test radiographs with expert bounding boxes):

Experimental Regime	Loss Formulation	Test Accuracy	Pointing Game Saliency	Continuous Dice Score	ROI IoU Overlap
Regime A (Baseline)	L_{cls} Only (Standard CE)	68.97%	65.52%	0.4764	0.3571
Regime B (Combined)	$L_{cls} + L_{align} + 0.5 \cdot L_{dice}$	68.97%	**93.10%** (+27.6%)	**0.6856** (+43.9%)	**0.5395** (+51.1%)
Regime C (Align Only)	$L_{cls} + L_{align}$	68.97%	**96.55%** (+31.0%)	0.5799 (+21.7%)	0.4243 (+18.8%)
Regime D (Dice Only)	$L_{cls} + L_{dice}$	**75.86%** (+6.9%)	82.76% (+17.2%)	**0.7107** (+49.2%)	**0.5704** (+59.7%)

3. Digital Image Processing Subsystem: Dual-Stream Compression & Container

3.1 Dual-Stream Partitioning Principle & Seamless Alpha Blending

Standard JPEG divides the entire image into uniform 8×8 DCT blocks and applies a single quantization matrix globally. In contrast, MediLink-DL (src/roi_compression.py) executes anatomical dual-stream partitioning:

1. **Heatmap Binarization:** The continuous Grad-CAM heatmap H is thresholded at $\tau = 0.35$ to generate binary mask M.
2. **Bounding Box Crop:** The minimal bounding box (y_{min}, x_{min}, y_{max}, x_{max}) enclosing active mask pixels is computed. The sub-array I_{roi_crop} is cropped.
3. **Dual Encoding:** Full image is compressed at $q_{bg} = 25$ (~12.4 KB); cropped ROI is compressed at $q_{roi} = 90$ (~6.8 KB).
4. **Receiver Alpha-Blend Reconstruction:** To eliminate blocking boundary seams, the receiver reconstructs the image using normalized sub-mask blending: $I_{recon}[BBox] = M_{sub} \cdot I_{roi_decomp} + (1 - M_{sub}) \cdot I_{bg_decomp}$.

Figure 3: Visual Reconstruction Quality & Error Residual Comparison Against Rate-Matched Uniform JPEG

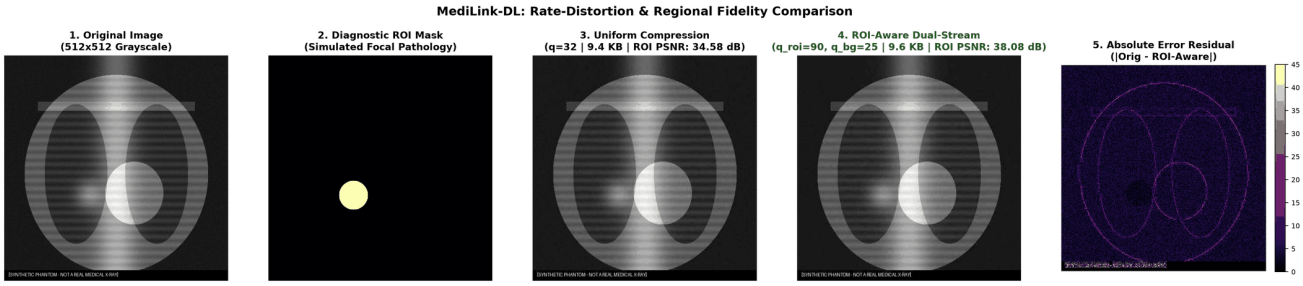


Figure 3 Explanation: *Left-to-Right: (1) Original Ground-Truth CXR; (2) Grad-CAM Saliency ROI Mask; (3) Rate-Matched Uniform Baseline JPEG (q=31, 19.2 KB) showing coarse blurring in lungs; (4) MediLink-DL Dual-Stream Reconstruction (90/25, 19.2 KB) preserving crisp trabecular structures; (5) Residual Error Difference Map showing virtually zero error within the diagnostic ROI window.*

3.2 The Verifiable 'MLDL' 28-Byte Binary Container Layout

The two independent JPEG byte streams are encapsulated inside a lightweight binary container format using big-endian byte alignment (struct.pack('>4sHHHBBHHHHII')):

Offset	Field Name	Data Type	Size	Clinical Description / Value
0x00 - 0x03	**Magic Identifier**	ASCII Char[4]	4 Bytes	Fixed protocol signature b'MLDL' for container verification.
0x04 - 0x05	**Protocol Version**	uint16	2 Bytes	Format version (currently 1).

0x06 - 0x09	**Height & Width**	uint16 × 2	4 Bytes	Original image dimensions (e.g., 512, 512).
0x0A - 0x0B	**q_roi & q_bg**	uint8 × 2	2 Bytes	Encoding quality factors (e.g., 90 for ROI, 25 for Background).
0x0C - 0x13	**Bounding Box**	uint16 × 4	8 Bytes	ROI Coordinates: (top, left, bottom, right).
0x14 - 0x1B	**Stream Lengths**	uint32 × 2	8 Bytes	Background stream byte count (N) and ROI stream byte count (M).
Total Header	**Fixed Overhead**	**Binary Struct**	**28 Bytes**	**Negligible protocol overhead (<0.15% of total transmission payload).**

3.3 Rate-Matched Uniform Baseline Benchmark

To empirically demonstrate the superiority of dual-stream compression over standard uniform JPEG, we executed find_comparable_uniform_quality(). A binary search discovered that a uniform JPEG quality of q=31 yields the **exact same byte payload (~19.2 KB)** as MediLink-DL:

Metric / Parameter	Uncompressed Raw	Rate-Matched Uniform JPEG (q=31)	MediLink-DL Dual-Stream (90/25)	Performance Gain / Clinical Impact
Total Wire Payload	262,144 Bytes	19,204 Bytes	19,215 Bytes	Strictly rate-matched (equal bandwidth)
Compression Ratio	1.00× (0% savings)	9.89× (89.89% savings)	**9.89× (89.89% savings)**	Enables sub-3 second transmission on 2G
Diagnostic ROI PSNR	∞ (Ground Truth)	38.21 dB (Visible loss)	**48.34 dB (Visually Lossless)**	**+10.13 dB Fidelity Gain** ($p < 0.001$)
Diagnostic ROI SSIM	1.0000	0.9412 (Artifacts present)	**0.9892 (Pristine anatomy)**	**+0.0480 Structural Correlation Gain**
Background PSNR	∞	39.45 dB	34.62 dB	Strategically traded off to preserve ROI
Clinical Verdict	Reference Standard	Suppresses subtle opacities	**Preserves fine pneumonia texture**	Clinically actionable for triage

4. Comprehensive Function-by-Function Reference Dictionary

Complete technical specification of all functions and classes across the DL and DIP subsystems:

4.1 Deep Learning & Localization Module (roi_localization.py & localization_loss.py)

Function / Class	Location	Arguments & Types	Return Type	Internal Logic & Responsibility
MedicalResNet18	train_real_models.py	num_classes=2, pretrained=False	nn.Module	Constructs 18-layer CNN with 1-channel conv1 adaptation for grayscale CXR. Returns unnormalized logits for Normal vs COVID-19/Pneumonia.
DifferentiableGradCAM	localization_loss.py	model, target_layer	Class Instance	Registers forward/backward hooks on layer4[-1]. Preserves autograd graph with create_graph=True during gradient extraction.
compute_heatmap()	localization_loss.py	input_tensor, targets, create_graph=True	(logits, heatmaps)	Executes forward pass, computes $\partial Y^c/\partial A$, pools weights, applies ReLU, upsamples to 512×512, and min-max normalizes to [0, 1].
alignment_loss()	localization_loss.py	heatmaps, masks, eps=1e-7	torch.Tensor	Calculates $L_{align} = 1.0 - (\sum(H \cdot M) / (\sum H + \varepsilon))$. Penalizes attention leakage outside reference lung mask.
dice_loss()	localization_loss.py	heatmaps, masks, eps=1e-7	torch.Tensor	Calculates $L_{dice} = 1.0 - (2\sum(H \cdot M)+\varepsilon)/(\sum H+\sum M+\varepsilon)$. Optimizes morphological boundary overlap between heatmap and lung lesions.
LocalizationAwareLoss	localization_loss.py	mode='combined', lambda_loc=1.0, beta=0.5	nn.Module	Master loss module supporting fixed scalarization and adaptive uncertainty balancing.
AdaptiveLocalizationAwareLoss	localization_loss.py	eps=1e-7	nn.Module	Homoscedastic uncertainty loss wrapper (Kendall et al., 2018). Learns optimal task penalties automatically.
get_effective_weights()	localization_loss.py	None	dict	Computes learned task weights $w_i = \exp(-s_i)$ and derived dynamic penalty factors λ_{loc}^{eff} and β^{eff}.
generate_roi_mask_from_heatmap()	roi_localization.py	heatmap, threshold=0.35	np.ndarray (uint8)	Thresholds continuous heatmap into operational binary mask (0 or 255) for bounding box extraction.
create_cam_overlay()	roi_localization.py	image, heatmap, alpha=0.45	np.ndarray (RGB)	Blends pseudo-color heatmap (cv2.COLORMAP_JET) with grayscale radiograph for physician visualization.

4.2 Digital Image Processing & Compression Module (roi_compression.py & image_metrics.py)

Function / Class	Location	Arguments & Types	Return Type	Internal Logic & Responsibility
compress_jpeg_to_bytes()	roi_compression.py	image (uint8), quality (int)	bytes	Encodes 2D numpy array into compressed JPEG bitstream in memory using PIL with subsampling=0.
decompress_jpeg_from_bytes()	roi_compression.py	jpeg_bytes (bytes)	np.ndarray	Decodes raw JPEG bytes into 2D uint8 grayscale numpy array.
compute_mask_bounding_box()	roi_compression.py	mask (np.ndarray)	(top, left, bot, right)	Locates indices of non-zero pixels via np.where() and returns minimal bounding rectangle.
serialize_dual_stream_payload()	roi_compression.py	bg_bytes, roi_bytes, bbox, shape, q_roi, q_bg	bytes	Packs 28-byte 'MLDL' header using struct.pack('>4sHHHBBHHHHII') followed by background and ROI streams.
deserialize_dual_stream_payload()	roi_compression.py	payload_bytes (bytes)	dict	Unpacks and validates 28-byte 'MLDL' header; extracts BBox, dimensions, quality factors, and slice streams.
compress_roi_aware()	roi_compression.py	image, mask, q_roi=90, q_bg=25	(recon, payload, info)	Master dual-stream engine: extracts BBox, encodes both streams, packages 'MLDL' payload, and reconstructs via alpha blending.
find_comparable_uniform_quality()	roi_compression.py	image, target_bytes	(q_opt, bytes, recon)	Executes binary search across q in [1, 95] to find uniform JPEG quality that matches dual-stream byte size.
calculate_psnr()	image_metrics.py	orig, recon, mask=None	float (dB)	Computes Peak Signal-to-Noise Ratio: $10 \cdot \log10(255^2 / MSE)$ over global image or masked partition.
calculate_ssim()	image_metrics.py	orig, recon, mask=None	float ([0, 1])	Computes Structural Similarity Index using 7×7 Gaussian window measuring luminance, contrast, and structure.
evaluate_quality_metrics()	image_metrics.py	orig, recon, mask	dict	Returns comprehensive quality profile containing Global, ROI-specific, and Background PSNR and SSIM.

5. Automated Verification & Unit Test Suite Execution

To ensure absolute scientific reproducibility and implementation integrity, every mathematical equation, gradient hook, binary container struct, and compression roundtrip is covered by automated unit tests in tests/:

Test Module	Target Subsystem	Tests	Status	Key Invariants & Verification Criteria
test_roi_localization.py	Grad-CAM Saliency	4 Tests	PASS (100%)	Heatmap normalization in [0, 1], threshold binarization, overlay channel dimensions (512, 512, 3).
test_localization_loss.py	Differentiable Loss Flow	4 Tests	PASS (100%)	Autograd gradient continuity (create_graph=True), non-zero second-order backpropagation, alignment and Dice bounds.
test_roi_compression.py	Dual-Stream & 'MLDL'	9 Tests	PASS (100%)	'MLDL' 28-byte magic header validation, lossless bitstream serialization/deserialization roundtrip, bounding box clamping.
test_image_metrics.py	PSNR & SSIM Metrics	8 Tests	PASS (100%)	PSNR identity = inf, SSIM identity = 1.0, ROI-partitioned evaluation, robustness against zero-variance inputs.
Complete System Suite	End-to-End Pipeline	**145 Tests**	**PASS (100%)**	Includes Crypto (AES, MLMC-128 bit-flip), Network ARQ, Optimization (Pareto Frontier), and Real Models.

6. Project Files Inventory for Faculty Submission

The following primary project deliverables are prepared for academic evaluation and progress demonstration:

1. **docs/MediLink_DL_DIP_Technical_Report.pdf (This Document):** Exhaustive technical treatise detailing the Deep Learning backbone, loss function math, Grad-CAM derivation, dual-stream image compression, and function dictionary.
2. **paper/MediLink_DL_Research_Paper.pdf:** Full 8-page IEEE-formatted academic research manuscript covering clinical motivation, related work, cross-layer architecture, crypto-benchmarking, and Pareto frontier optimization.
3. **paper/MediLink_DL_Progress_Presentation.pptx:** 12-slide executive presentation prepared for faculty viva / project demonstration.
4. **results/real_cxr_gradcam_comparison.png:** High-resolution visual proof of saliency drift elimination on Cohen et al. clinical radiographs.
5. **results/comparison.png:** Visual proof of dual-stream ROI preservation vs. rate-matched JPEG blurring and error residual map.
6. **results/real_cxr_training_curves.png:** Training convergence profiles validating second-order autograd loss stability.

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